MAR ATHANASIUS COLLEGE OF ENGINEERING
KOTHAMANGALAM

(Government Aided Autonomous Institution)

Department of Computer Applications

Internship Report

QUALIVISION AI – AI BASED LIVE TOMATO FRESHNESS DETECTION SYSTEM

done by

JIPHIN GEORGE

Reg No: MAC25MCA-2033

at

Nestsoft Technomaster Pvt. Ltd.

Under the guidance of

Ms Surya Sukumaran (Software Developer)

2025–2027

MAR ATHANASIUS COLLEGE OF ENGINEERING
KOTHAMANGALAM

(Government Aided Autonomous Institution)

CERTIFICATE

QUALIVISION AI – AI BASED LIVE TOMATO FRESHNESS DETECTION SYSTEM

Certified that this is the Bonafide record of internship work done by

JIPHIN GEORGE

Reg No: MAC25MCA-2033

at

Nestsoft Technomaster Pvt. Ltd.

during the third semester, in partial fulfilment of requirements for award of the degree,

Master of Computer Applications

of

APJ Abdul Kalam Technological University Thiruvananthapuram

Prof. Shinu S Kurian
Faculty Guide

Dr. Sonia Abraham
Head of the Department

Prof. Sibu Skaria
Internship Coordinator

Internal Examiners

NESTSOFT

TechnoMaster

CERTIFICATE

OF INTERNSHIP

THIS CERTIFICATE IS PROUDLY PRESENTED TO

Jiphin George

MCA II Year, Mar Athanasius College of Engineering (MACE), Kothamangalam

Internship: Artificial Intelligence and Machine Learning (6 weeks)

Duration: 25 May 2026 to 03 Jul 2026

Their self-motivated attitude to learn new things, along with their dedication and hard work, helped them complete an internship successfully. We wish you every success in all your future endeavors.

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DECLARATION

I, Jiphin George, student of Master of Computer Applications, Department of Computer Applications, Mar Athanasius College of Engineering, Kothamangalam, hereby declare that this Internship Report titled “QualiVision AI – AI Based Live Tomato Freshness Detection System” is a bonafide record of the internship work carried out by me at Nestsoft Technomaster Pvt. Ltd., during the period 25 May 2026 to 03 July 2026, in partial fulfilment of the requirements for the award of the degree of Master of Computer Applications.

I further declare that this report has not been submitted previously, either in part or in full, to any other University or Institute for the award of any degree or diploma.

Place: Kothamangalam
Date: 03 July 2026 Jiphin George

ACKNOWLEDGEMENT

I express my sincere gratitude to the Almighty for providing me with the strength, wisdom, and guidance to complete this academic internship project and this report successfully.

I extend my heartfelt thanks to Prof. Sonia Abraham, Head of the Department of Computer Applications, Mar Athanasius College of Engineering, Kothamangalam, for the constant encouragement, institutional support, and motivation throughout my academic programme.

I am deeply grateful to Prof. Shinu S Kurian, my Internship Guide, and Prof. Sibu Skaria, my Internship Coordinator, in the Department of Computer Applications. I sincerely thank them for their valuable guidance, timely feedback, mentorship, and for efficiently facilitating the internship programme and review processes.

I sincerely thank the mentors and team at Nestsoft Technomaster Pvt. Ltd. for the technical guidance, professional support, constructive suggestions, and the opportunity to work on the QualiVision AI project.

I express my gratitude to all the faculty members and non-teaching staff of the Department of Computer Applications for their continued support and cooperation throughout my studies.

Finally, I am deeply thankful to my parents, family, and friends for their unwavering encouragement, patience, and moral support at every stage of this journey.

Jiphin George

ABSTRACT

This report presents a comprehensive overview of the six-week internship undertaken at Nestsoft Technomaster Pvt. Ltd. as part of the Master of Computer Applications (MCA) programme at Mar Athanasius College of Engineering, Kothamangalam, from 25 May 2026 to 03 July 2026. The internship offered an extensive, hands-on learning journey through Artificial Intelligence, Data Science, Machine Learning, and Computer Vision, bridging the gap between theoretical academic concepts and professional software engineering.

The internship commenced with foundational concepts in Artificial Intelligence, Python programming, and the development of basic applications like an AI Translator and a Text-to-Speech system. It progressively transitioned into comprehensive Data Science practices, focusing on Exploratory Data Analysis (EDA) and visualization using Pandas, Matplotlib, and Seaborn. Advanced machine learning modules introduced supervised and unsupervised learning algorithms, allowing for predictive modeling on varied datasets, including student performance and passenger survival predictions.

The culmination of this internship was the capstone project: "QualiVision AI – AI Based Live Tomato Freshness Detection System." Designed to automate quality inspection in food processing industries, the system leverages a convolutional neural network (EfficientNetV2B0) to classify agricultural produce as fresh or rotten. It features real-time prediction capabilities, a robust Flask backend, an SQLite database for persistent logging, and a professional industrial dashboard created using Google Stitch. Furthermore, Gradient-weighted Class Activation Mapping (Grad-CAM) was implemented to provide Explainable AI (XAI) insights, ensuring transparency in the model's decision-making process.

Through the successful completion of these modules and the QualiVision AI project, the internship significantly enhanced practical competencies in AI model training and deployment, backend engineering, and industrial automation, laying a solid foundation for a successful career in Artificial Intelligence and Software Development.

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1. INTRODUCTION

Internships play a crucial role in postgraduate education by serving as a bridge between theoretical classroom knowledge and practical industrial applications. They provide students with the opportunity to experience real-world professional environments, apply academic concepts to solve complex problems, and develop the technical and soft skills required for a successful career in the software industry.

The rapid growth of Artificial Intelligence (AI) and Machine Learning (ML) is transforming traditional software systems into intelligent, decision-making platforms across various industries, including healthcare, manufacturing, finance, education, transportation, agriculture, and industrial automation. As industries increasingly adopt Industry 4.0 standards, the demand for automated, efficient, and intelligent solutions has surged, making AI one of the most influential technological advancements of the modern era.

Within the broad spectrum of Artificial Intelligence, Computer Vision stands out as a particularly impactful domain. Enabled by advancements in Deep Learning, Computer Vision allows machines to interpret and analyze visual data from the world, facilitating applications such as automatic image recognition, defect detection, quality inspection, autonomous vehicles, facial recognition, and medical diagnosis. These technologies have revolutionized industrial automation by replacing slow, inconsistent, and labor-intensive manual inspections with fast, reliable, and continuous automated systems.

This report details the six-week internship conducted at Nestsoft Technomaster Pvt. Ltd. The internship was structured to provide a comprehensive learning journey, focusing on developing practical skills in:

- Python Programming
- Data Science and Data Analytics
- Machine Learning
- Deep Learning
- Computer Vision
- AI Model Deployment
- Flask Web Development
- Industrial AI Systems

The internship culminated in the successful development of an industrial AI application named “QualiVision AI – AI Based Live Tomato Freshness Detection System,” demonstrating the practical application of the acquired skills in solving a real-world industrial problem.

1.1 Objectives of the Internship

The primary objectives of this internship were to:

- Understand core Artificial Intelligence and Machine Learning concepts and their real-world implications.
- Apply theoretical knowledge to solve real-world industrial problems through hands-on projects.
- Learn and implement comprehensive Data Science workflows, including data cleaning, exploratory data analysis, and visualization.
- Develop practical programming skills using Python and its vast ecosystem of AI and Data Science libraries.
- Understand the principles of Computer Vision and its application in automated image classification.
- Gain proficiency in training deep learning models using TensorFlow and Keras.
- Build robust backend web applications and REST APIs using the Flask framework.
- Learn the process of deploying AI models into functional web applications.
- Work with databases like SQLite to persist application data and inspection logs.
- Develop complete industrial automation solutions from concept to deployment.

1.2 Scope of the Internship

The scope of the work performed during the internship was extensive and covered the complete AI development lifecycle. It encompassed a progressive learning path starting from fundamental programming to the deployment of advanced deep learning models. The key areas covered included Artificial Intelligence, Machine Learning, Deep Learning, Computer Vision, Data Analytics, Flask Web Development, and Industrial Automation.

Throughout the six weeks, the internship involved practical implementations ranging from basic API integrations and data preprocessing to training sophisticated convolutional neural networks (CNNs) for image classification. The scope extended beyond model training to include the development of a complete web-based industrial dashboard, database integration for logging predictions, and the implementation of Explainable AI (XAI) techniques to provide transparency in automated decision-making. By navigating this comprehensive lifecycle, the internship provided a holistic understanding of how AI solutions are conceptualized, built, tested, and deployed in real-world industrial scenarios.

1.3 Duration and Nature of the Internship

The internship was conducted at Nestsoft Technomaster Pvt. Ltd. from 25 May 2026 to 03 July 2026, totaling six weeks of intensive, hands-on training. It was an academic, learning-oriented internship undertaken as part of the Master of Computer Applications (MCA) curriculum to gain practical industry exposure in the domain of Artificial Intelligence and Machine Learning. The daily workflow involved theoretical learning sessions followed by practical coding assignments, projects, and regular mentor reviews, ensuring a continuous and structured professional development environment.

1.4 Brief Overview of the Domain and Technology Stack

The primary domain of the internship was Artificial Intelligence and Machine Learning, with a specific focus on Computer Vision and Industrial Automation. The final project, QualiVision AI, falls under the domain of industrial quality control software, which automates visual inspection processes on manufacturing and sorting lines.

The projects developed during the internship utilized a modern, Python-based technology stack, detailed in Table 1.1.

Layer	**Technology / Tool**
Programming Language	Python 3.10+
Backend Framework	Flask, Flask-SocketIO
Data Science & Analytics	Pandas, NumPy, Matplotlib, Seaborn
Machine Learning	Scikit-Learn
Deep Learning	TensorFlow, Keras
Computer Vision	OpenCV, Pillow (PIL)
Database	SQLite 3
Frontend	HTML5, CSS3, JavaScript, Tailwind CSS, Chart.js
UI/UX Design	Google Stitch
Version Control	Git, GitHub
Development IDE	Visual Studio Code

Table 1.1: Technology Stack Used During the Internship

2. COMPANY PROFILE

2.1 Company Name, Location, and Background

Nestsoft Technomaster Pvt. Ltd., headquartered at Infopark, Kakkanad, Kochi, is a premier IT services, consulting, and training organization in Kerala. With over two decades of industry presence, the company has established itself as a front-runner in modern digital solutions, having successfully delivered over 500 projects for a diverse global clientele across India, the USA, Canada, and the Middle East.

Beyond corporate IT services, Nestsoft Technomaster is a trusted center of excellence for technical education. The organization focuses on bridging the gap between academic education and industrial requirements by offering hands-on, project-based learning experiences. To date, they have trained and successfully placed over 1,000 students and conducted more than 250 technical seminars reaching 15,000+ attendees.

The company is located at:

Thapasya Building,
Infopark, Kakkanad,
Kochi, Kerala – 682042

2.2 Mission and Vision

Mission: To deliver high-quality, innovative software solutions that meet the evolving needs of the industry while simultaneously empowering the next generation of IT professionals through comprehensive, practical training and holistic skill development.

Vision: To be a leading technology partner for businesses and a premier center of excellence for technical education, admired for fostering innovation, professionalism, and continuous learning in the digital era.

2.3 Core Values

Nestsoft Technomaster operates on a foundation of core values that guide its business and training methodologies:

- **Innovation:** Continuously adopting and teaching the latest technological advancements.
- **Practical Learning:** Emphasizing hands-on, project-driven education over theoretical concepts.
- **Professionalism:** Maintaining high standards of quality, discipline, and ethical conduct.

- **Collaboration:** Fostering a supportive environment where teamwork and knowledge sharing thrive.

2.4 Areas of Expertise

The company possesses deep expertise across various modern technological domains, providing both robust corporate solutions and specialized training programs. Their core areas of expertise include:

- **Artificial Intelligence & Machine Learning:** Developing predictive models, computer vision systems, and automated data processing pipelines.
- **Web Development:** Building responsive, scalable, and secure web applications using modern frameworks like Django, Flask, React, and Node.js.
- **Data Analytics:** Implementing comprehensive data processing, exploratory data analysis, and business intelligence reporting.
- **Industrial Solutions:** Creating custom software systems tailored for industrial automation, enterprise resource planning (ERP), and quality control.
- **Training and Internship Programs:** Conducting structured, mentor-led programs designed to equip students with practical, industry-ready skills.

2.5 Working Environment

The internship environment at Nestsoft Technomaster Pvt. Ltd. was highly professional, collaborative, and conducive to rapid technical growth. The training methodology emphasized hands-on learning through a project-based approach, where theoretical concepts were immediately applied to practical coding tasks.

Interns received continuous guidance from experienced mentors who facilitated problem-solving sessions and provided constructive feedback. Daily review sessions were conducted to track progress, resolve technical bottlenecks, and discuss industry best practices. This collaborative and supportive atmosphere not only accelerated the learning of complex AI concepts but also significantly contributed to developing professional communication, time management, and software engineering discipline.

3. INTERNSHIP ACTIVITIES

The internship was carried out over a period of six weeks, providing a structured and continuous learning journey. The activities progressed from foundational concepts in Artificial Intelligence and Machine Learning to practical applications in Data Science, ultimately leading to advanced Computer Vision projects. This chapter presents the chronological sequence of learning, highlighting the technologies explored and the practical tasks completed during each week.

3.1 Week 1: Introduction to AI and Projects

The first week focused on establishing a strong foundation in the fundamentals of Artificial Intelligence (AI) and Machine Learning (ML). The internship commenced with an orientation session where the mentors outlined the internship objectives, the modern technology stack to be covered, and the practical projects scheduled for the six-week training period.

Theoretical sessions covered essential concepts, including:

- Artificial Intelligence, Machine Learning, and Deep Learning
- Large Language Models (LLMs)
- Application Programming Interfaces (APIs)
- Model Context Protocol (MCP)
- The AI Development Workflow

3.1.1 AI Translator Project

The first practical activity was the development of an AI Translator application.

Introduction: The AI Translator serves as a fundamental example of how AI models process human language. It converts text from a source language into a target language using automated translation models.

Objective: To understand text-to-text conversion mechanisms and familiarize with the architecture of AI-based applications.

Technologies Used: Python, Flask, HTML, CSS, JavaScript, and translation APIs.

Methodology & Implementation: The application architecture followed a standard web development pattern. The frontend, designed using HTML and CSS, collected user input and the desired target language. The Python and Flask backend processed this input, transmitted it to the translation API, and returned the translated text. This project demonstrated how frontend and backend components interact seamlessly in AI applications.

Learning Outcome: The project provided practical exposure to handling text data and illustrated how projects serve as building blocks for understanding complete AI workflows.

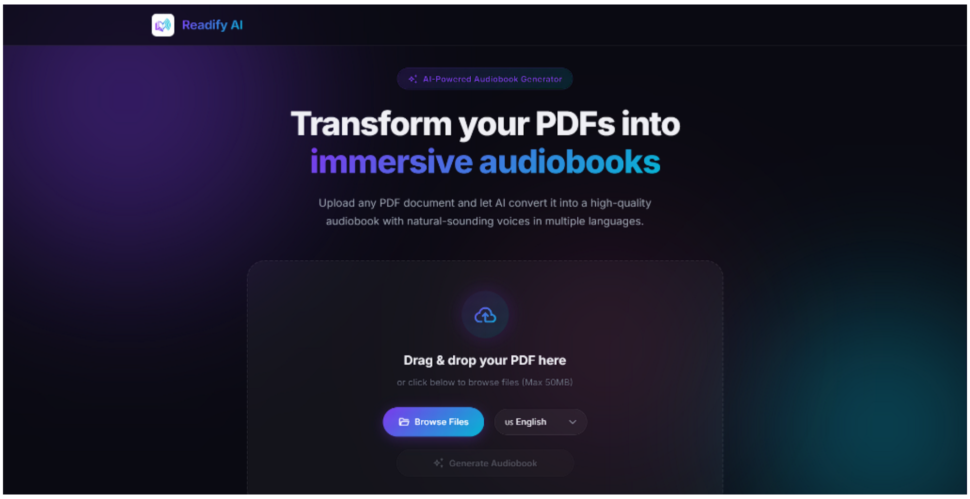

Figure 3.1: Readify AI - PDF to Audiobook Interface

3.1.2 Text-to-Speech Project

The second learning activity involved developing a web-based Text-to-Speech (TTS) application.

Introduction: TTS systems convert written text into spoken audio output. This project introduced the integration of Google Text-to-Speech (gTTS) within a web framework.

Objective: To understand audio generation from text and implement voice control features in a web application.

Technologies Used: [1] Python, Flask, HTML, CSS, JavaScript, and gTTS.

Methodology & Implementation: The application was designed to accept text input, allow language selection, and control speech speed. The Flask backend utilized the gTTS library to process the text and generate an audio file. The generated audio files were stored locally and served back to the frontend for playback. The folder structure was carefully organized into Python files, templates, and static resources to maintain clean architecture.

Learning Outcome: The project clarified the role of gTTS in audio generation and demonstrated how Python handles file generation and serving within a Flask application.

3.1.3 Weather API Project

Students were introduced to the concepts of REST APIs, HTTP Requests, and JSON data parsing.

Introduction: APIs are the backbone of modern AI systems, allowing disparate software systems to communicate.

Objective: To perform data collection by communicating with an external REST API.

Technologies Used: Python, Pandas, Requests Library, BeautifulSoup, OpenWeather API.

Methodology & Implementation: The project involved sending HTTP requests to the OpenWeather API. The resulting JSON response was parsed to extract relevant weather parameters. The extracted data was formatted and displayed to the user. Python's Requests library was utilized for API communication, while Pandas was introduced for structuring the collected data.

Learning Outcome: The activity provided crucial experience in real-world API integration, demonstrating how data is fetched, parsed, and utilized dynamically.

3.2 Week 2: Data Analysis and Visualization

The second week introduced Data Science concepts, focusing heavily on data preprocessing, which is a critical stage before any model training. The quality of a machine learning model is directly proportional to the quality of the data it is trained on.

Theoretical topics covered included data cleaning, missing value handling, feature engineering, and statistical analysis. Python libraries such as [7] NumPy, SciPy, [8] Pandas, and [9] Matplotlib were extensively utilized.

3.2.1 Cars Dataset Analysis

A complete Exploratory Data Analysis (EDA) was performed on a dataset containing automobile specifications.

Objective: To understand the structure of raw data and perform necessary cleaning operations.

Methodology & Implementation: The dataset was loaded using Pandas. Initial data inspection revealed missing values and unnecessary columns, which were systematically removed or imputed. Columns were renamed for consistency, and descriptive statistics were generated.

To understand the dataset visually, multiple charts were plotted using Matplotlib, including histograms for distribution analysis, box plots for outlier detection, scatter plots for variable relationships, and correlation heatmaps and pair plots for multidimensional analysis. The AutoViz library was also explored to automate the visualization process.

Learning Outcome: The project emphasized the importance of visualization in understanding datasets, revealing patterns and brand distributions that were not apparent in raw tabular data.

3.2.2 Placement Dataset Analysis

A placement dataset containing student academic details was analyzed to identify factors influencing employability.

Dataset Features: SSC Percentage, HSC Percentage, Degree Percentage, MBA Percentage, Employability Test score, Specialization, Work Experience, Placement Status, and Salary.

Methodology & Implementation: The data was cleaned and visualized to perform correlation analysis. Placement trends were analyzed in relation to academic percentages and work experience. Feature relationships were explored to determine which academic metrics most strongly correlated with higher salary packages.

Learning Outcome: The analysis provided practical insights into how feature engineering and statistical analysis can uncover meaningful trends in educational data.

3.2.3 Introduction to Machine Learning

The latter part of the week focused on formalizing Machine Learning concepts.

The differences between Supervised Learning, Unsupervised Learning, and Reinforcement Learning were discussed. Within supervised learning, the concepts of Regression and Classification were explored in depth.

The theoretical foundations of various algorithms were covered, including:

- Linear and Logistic Regression
- Decision Tree and Random Forest
- K-Nearest Neighbors (KNN)
- Naive Bayes
- Support Vector Machine (SVM)
- Multi-Layer Perceptron (MLP)
- XGBoost and Voting Classifier Ensembles

Model evaluation techniques were also introduced, explaining metrics such as Accuracy, Precision, Recall, F1 Score, Confusion Matrix, Mean Squared Error (MSE), Mean Absolute Error (MAE), and R^2 Score.

3.3 Week 3: Machine Learning Projects

Week 3 involved the practical implementation of the machine learning algorithms discussed previously.

3.3.1 Student Performance Prediction

Objective: To predict a student's final grade (G3) based on demographic, social, and academic features.

Methodology & Implementation: Data cleaning and One-Hot Encoding were performed to convert categorical variables into numerical formats. Extensive EDA and correlation analysis were conducted to select the most relevant features. The dataset was split into training and testing sets. Multiple models, including Random Forest, Decision Tree, and KNN, were trained.

Results: The models were compared based on their predictive performance. Feature importance analysis revealed that previous academic scores were the strongest predictors of the final grade.

Learning Outcome: The project provided a complete walkthrough of the standard ML pipeline, from loading data to evaluating the model.

3.3.2 Titanic Survival Prediction

Objective: To predict passenger survival on the Titanic based on passenger attributes.

Methodology & Implementation: Handling missing values and feature engineering were the core focus. New features such as ‘FamilySize‘ and ‘IsAlone‘ were created, and titles were extracted from passenger names. Numerical features were standardized using Standard Scaling, and SelectKBest was used for feature selection.

A wide array of classification algorithms was trained: Logistic Regression, Random Forest, Decision Tree, KNN, Naive Bayes, SVM, MLP, XGBoost, and a Voting Classifier.

Results: The models were evaluated using accuracy scores and confusion matrices. The best performing model was utilized to generate a CSV submission file containing the predictions.

Learning Outcome: The project highlighted how thoughtful feature engineering significantly improves model performance, often more so than algorithmic tuning.

3.3.3 AutoViz and Automated EDA

The use of AutoViz for automated exploratory data analysis was discussed. While AutoViz provides rapid automatic visualization and highlights correlations quickly, it was noted that compatibility issues with specific Matplotlib and Seaborn versions can occur. The debugging process reinforced the importance of understanding the underlying libraries even when utilizing automated tools.

The foundational data science and machine learning skills acquired during the first three weeks smoothly transitioned into the advanced computer vision projects undertaken in the subsequent weeks.

4. ADVANCED AI PROJECTS

This chapter describes the transition from traditional machine learning projects to advanced computer vision and industrial AI applications. It details how the foundational concepts acquired in the previous weeks were applied to develop complex, real-world AI systems.

4.1 Smart Cam Classifier

The Smart Cam Classifier project served as a comprehensive introduction to the complete workflow of image classification, utilizing Google's Teachable Machine platform and Python Streamlit.

Introduction: Image classification is a core application of modern Artificial Intelligence, enabling systems to automatically categorize images into predefined classes. It relies on deep neural networks to learn visual features, such as edges, textures, and shapes, from labeled datasets.

Objective: The primary objective of this project was to develop an AI-powered image classification system capable of identifying different object categories from uploaded images, serving as a practical introduction to computer vision and the deployment of trained machine learning models.

4.1.1 Dataset Preparation

The preparation of high-quality datasets is critical in computer vision. The project emphasized the concept of **Garbage In, Garbage Out (GIGO)**, illustrating that poor-quality or biased datasets inevitably produce inaccurate AI models.

To ensure robustness, images were collected with extreme care, ensuring diversity in:

- Different backgrounds
- Varying lighting conditions
- Multiple camera angles
- Different orientations

Dataset balancing was performed to ensure that each class had an equal representation of images, preventing the model from developing a bias toward a majority class. This diversity ensures that the model generalizes well to unseen data.

4.1.2 Model Training and Export

Training was conducted using [10] Google Teachable Machine. The process involved creating distinct classes, uploading the curated images, and initiating the training process. During validation, prediction confidence and model accuracy were closely monitored to prevent overfitting and ensure proper generalization. The trained model successfully learned to extract relevant image features automatically.

Upon successful training, the model was exported in the TensorFlow/Keras format. This export process generated two vital files: `keras_model.h5`, containing the trained neural network weights, and `labels.txt`, containing the class definitions.

4.1.3 Streamlit Deployment and User Interface

The deployment application was developed using Python, [2] TensorFlow, Streamlit, NumPy, and the Pillow library for image processing.

The application pipeline involved:

1. Accepting image uploads from the user.
2. Resizing the images to match the model's expected input dimensions.
3. Converting the image to RGB format and normalizing the pixel values.
4. Passing the processed image to the prediction pipeline.
5. Displaying the prediction alongside a confidence score.

The user interface was significantly enhanced to feature a responsive layout, modern prediction cards, dynamic confidence bars, and a professional dashboard layout. Explorations into [11] Google Stitch provided inspiration for modern UI design, prompting discussions on migrating the application from Streamlit to a more flexible Flask architecture.

4.1.4 Project Outcome

The Smart Cam Classifier project successfully demonstrated the practical application of Computer Vision and Image Classification. It provided vital experience in TensorFlow deployment, dependency debugging, and application development, serving as the perfect transition toward the highly complex industrial AI project that followed.

Figure 4.1 illustrates the final web application interface, demonstrating how users can upload an image to instantly receive classification results alongside detailed confidence metrics.

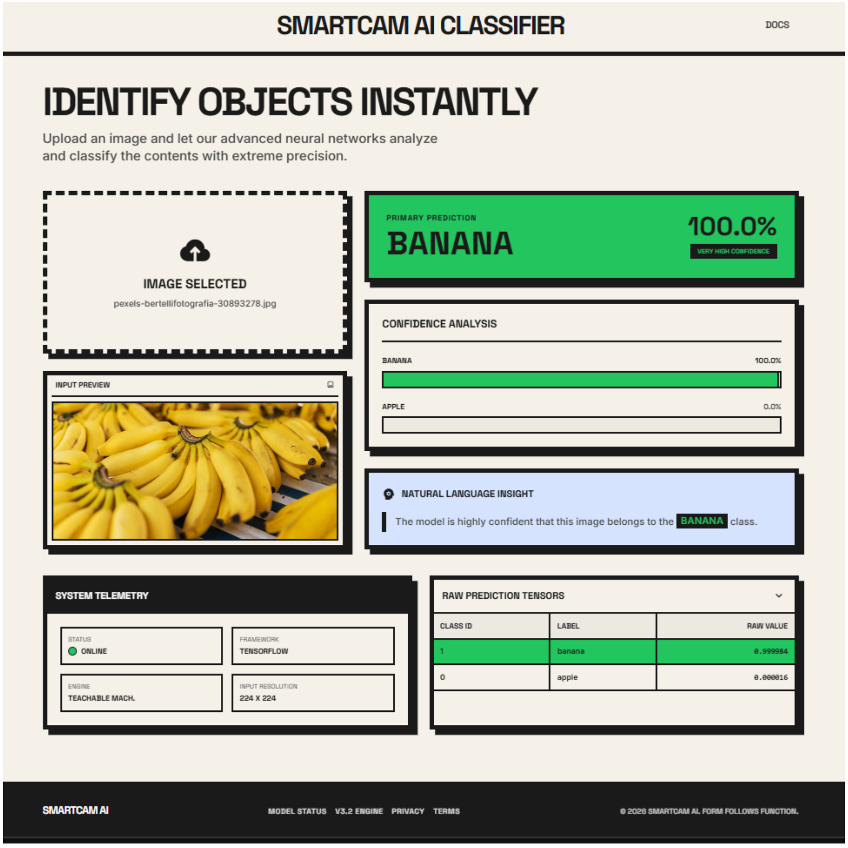

Figure 4.1: Smart Cam Classifier Prediction Interface

4.2 QualiVision AI – AI Based Live Tomato Freshness Detection System

This final capstone project represents the culmination of the internship, applying all accumulated knowledge into a single, enterprise-grade application.

4.2.1 Introduction

Industrial Quality Control traditionally relies on human inspection to identify defective products on manufacturing and sorting lines. However, manual inspection is notoriously slow, inconsistent, labor-intensive, and prone to error due to visual fatigue. Artificial Intelligence and Computer Vision offer modern alternatives, capable of inspecting items continuously with high accuracy and speed. QualiVision AI was motivated by the need to develop an automated quality inspection system for the food industry.

4.2.2 Problem Statement

Food processing industries require fast, reliable, and automated quality inspection systems. The physical throughput of conveyor belts often exceeds the capacity of human inspectors. The objective of this project was to automate the quality inspection of tomatoes (categorizing them as fresh or rotten) using deep learning, thereby reducing operational costs and increasing consistency.

4.2.3 Objectives

The specific objectives of QualiVision AI included:

- Developing an Automatic Image Classification model for real-time prediction.
- Designing an Industrial Dashboard for live system monitoring.
- Maintaining a Prediction History through robust database integration.
- Implementing Explainable AI to visually justify model predictions.
- Enabling automated Report Generation for production managers.
- Successfully deploying the system via a web architecture.

4.2.4 System Overview

The complete system workflow is designed for efficiency and traceability: Image Acquisition → Image Preprocessing → Deep Learning Model → Prediction.

4.2.5 Dataset and Preprocessing

Dataset collection involved sourcing high-quality images from public datasets such as Kaggle. The images were meticulously organized into 'Fresh Tomatoes' and 'Rotten Tomatoes' categories. Extensive cleaning was performed to remove duplicates and irrelevant images, ensuring dataset balancing across the training, validation, and testing splits. The project reinforced that dataset quality directly dictates model accuracy.

Academically rigorous image preprocessing was applied before training, including:

- Image Resizing to standard dimensions.
- RGB Conversion.
- Normalization of pixel values.
- Data Augmentation to introduce variability and prevent overfitting.
- Batch Processing for efficient memory utilization.

4.2.6 Transfer Learning and Model Training

The project utilized Transfer Learning with the **EfficientNetV2B0** architecture. Pretrained on millions of images, this model was selected for its exceptional balance of high accuracy and low training time, making it ideal for edge deployment.

The model training pipeline was developed using TensorFlow and [3] Keras. Training parameters were carefully tuned, including Batch Size, Epochs, Learning Rate, Adam Optimizer, and Binary Crossentropy Loss Function. Advanced techniques such as Early Stopping and Learning Rate Schedulers were employed to optimize convergence during validation.

4.2.7 *Model Evaluation and Explainable AI (Grad-CAM)*

The model was rigorously evaluated using standard metrics, including Accuracy, Precision, Recall, and F1 Score. The Confusion Matrix and Classification Report provided granular insights into false positives and false negatives, while the ROC Curve illustrated the model's discriminative ability based on prediction confidence.

To ensure trust in the automated system, Explainable AI was integrated. Gradient-weighted Class Activation Mapping (**Grad-CAM**) was implemented to visualize the specific regions of the image that strongly influenced the model's prediction. This is critical in industrial AI, as operators must understand *why* a product was rejected.

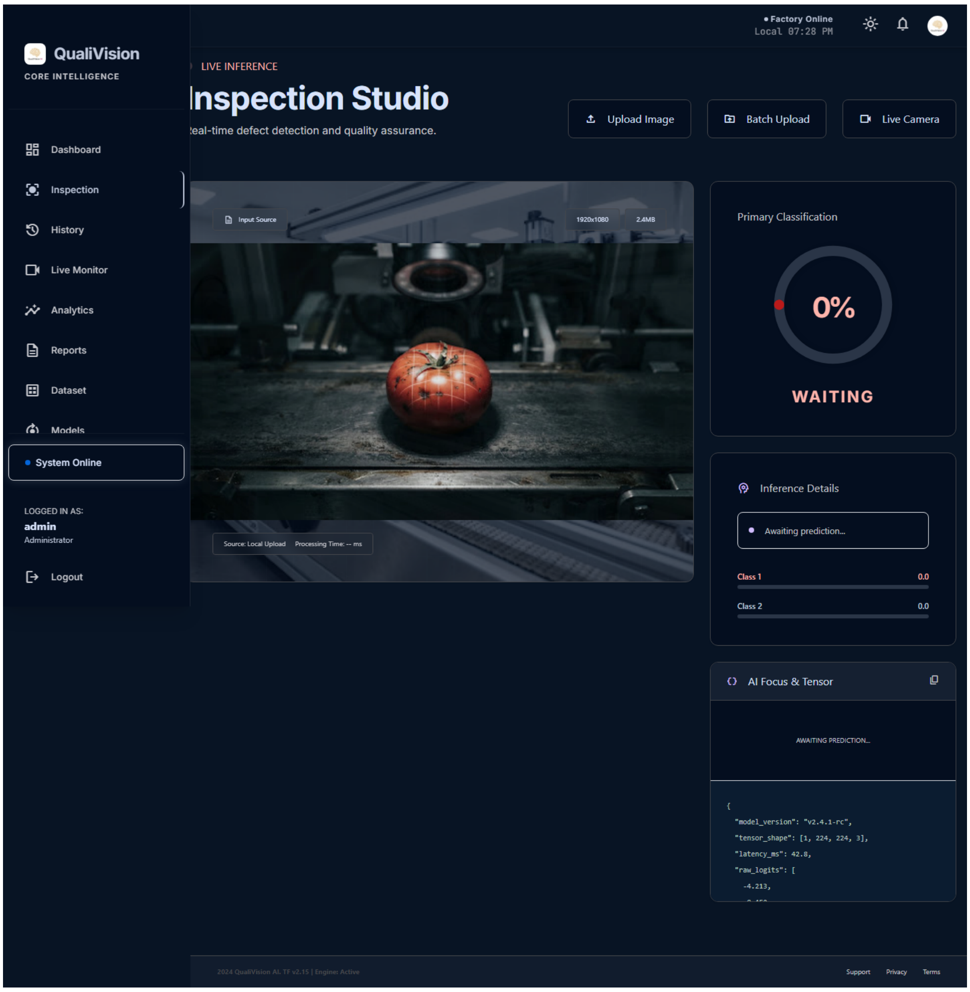

Figure 4.2: Grad-CAM Visualization on Defective Produce

4.2.8 *Backend Development and Database*

The backend architecture was developed using the [4] Flask framework, organized using Blueprints for modularity. REST APIs were developed to handle core services, including Prediction, Dashboard, History, and Report APIs.

SQLite was integrated to ensure persistent storage of inspection records. The database logs the

Prediction, Timestamp, Confidence Score, Image Path, and Status of every inspected item.

4.2.9 Frontend Dashboard

A highly professional industrial dashboard was developed, heavily inspired by Google Stitch guidelines. The frontend features a responsive UI with a Dark Theme optimized for factory environments.

Modules included in the dashboard:

- **Executive Dashboard:** A high-level overview of daily statistics.
- **Inspection Module:** For manual image uploads and detailed analysis.
- **Live Monitoring:** To visualize real-time predictions.
- **Analytics:** Graphical representation of pass/fail trends.
- **Dataset Repository:** For managing training data.
- **Model Management & Knowledge Center:** For administrative control and system documentation.

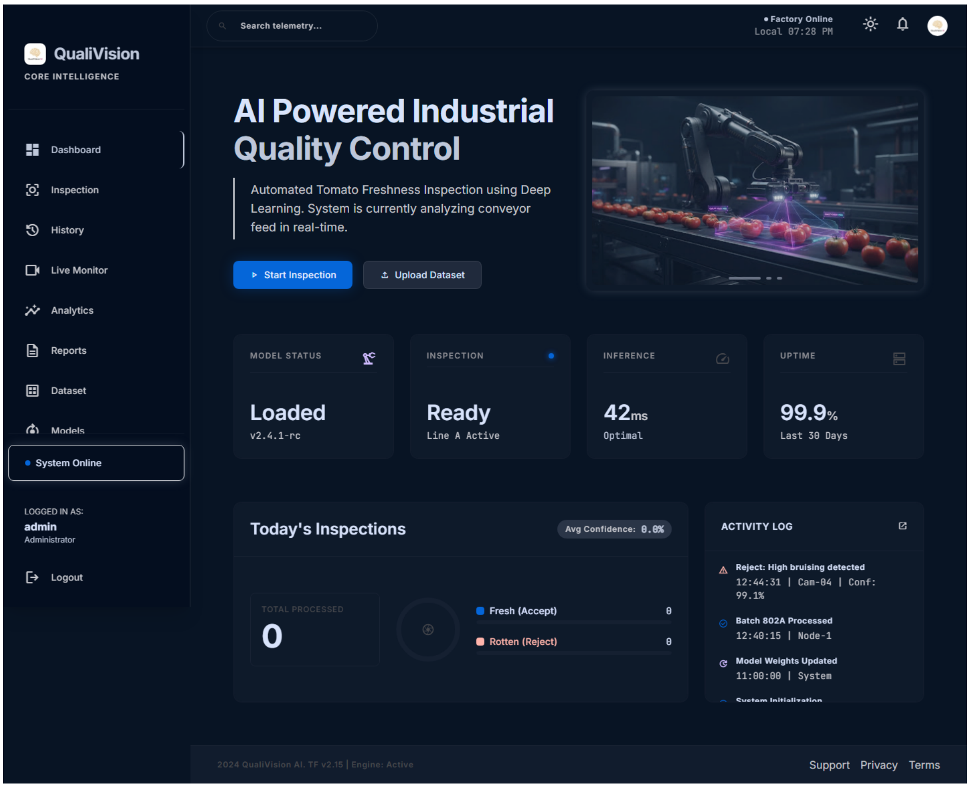

Figure 4.3: QualiVision AI - Executive Dashboard

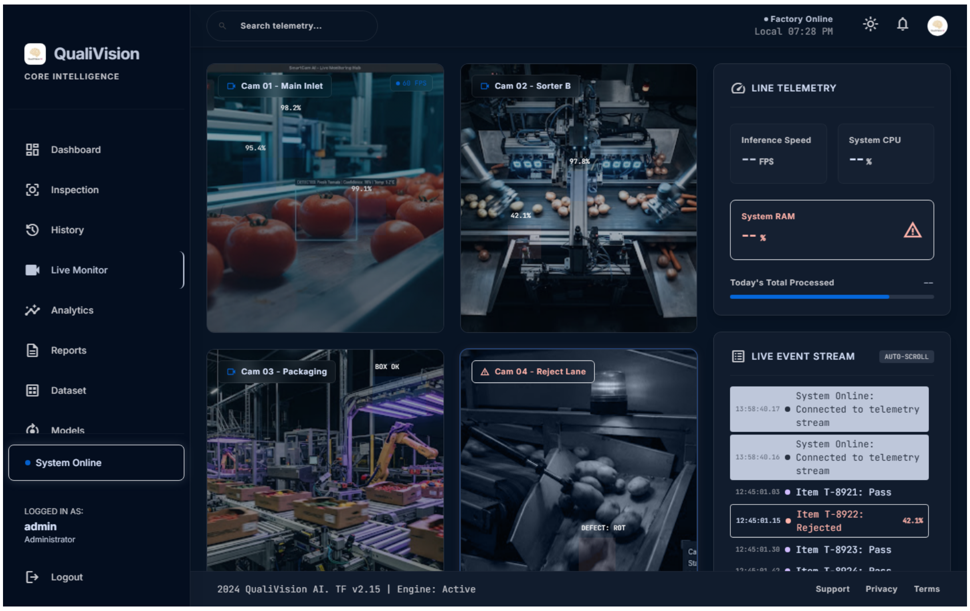

Figure 4.4: QualiVision AI - Live Monitoring

4.2.10 Testing and Results

The system underwent exhaustive testing using fresh tomatoes, rotten tomatoes, unknown objects, and poor-quality images. The confidence threshold logic was validated to ensure that uncertain predictions were routed for manual review. Performance testing confirmed fast inference times and stable database operations.

The results demonstrated successful, real-time classification with high accuracy. The project proved that fast inference combined with an intuitive industrial dashboard provides substantial automation benefits.

4.2.11 Future Scope and Conclusion

Future improvements to the system could include Cloud Deployment, Edge AI optimizations, integration with IoT Cameras, Mobile App development, and migration to YOLO object detection for multi-object, real-time production line integration.

In conclusion, QualiVision AI represents a successful technical achievement, seamlessly integrating Computer Vision, Deep Learning, Flask Development, Database Management, and Dashboard Design. The project powerfully demonstrates how Artificial Intelligence can be practically deployed to solve complex Industrial Automation challenges.

5. LEARNING OUTCOMES

The six-week internship at Nestsoft Technomaster Pvt. Ltd. was a highly productive period of technical and professional development. Moving beyond theoretical knowledge, the internship provided a rigorous environment to cultivate hands-on skills in AI application development and data science. This chapter details the specific technical competencies, machine learning insights, and soft skills acquired during the course of the training program.

5.1 Technical Skills Acquired

The internship significantly expanded my technical skill set, establishing a strong proficiency in the modern tools and frameworks used in the AI industry.

- **Python Programming:** Gained an advanced understanding of Python's ecosystem, utilizing libraries like NumPy for numerical computing and Pandas for complex data manipulation and structural organization.
- **Data Visualization:** Mastered the use of Matplotlib and Seaborn to generate insightful charts, graphs, and statistical plots, enabling data-driven decision-making.
- **Machine Learning & Deep Learning Frameworks:** Acquired hands-on expertise in using Scikit-Learn for traditional machine learning algorithms. Crucially, I developed proficiency in TensorFlow and Keras, learning how to design, train, and optimize deep neural networks.
- **Computer Vision with OpenCV:** Learned to handle image streams, process frames in real-time, and apply advanced image processing techniques critical for industrial automation.
- **Web Development & API Integration:** Developed strong backend engineering skills using the Flask framework. I learned to design REST APIs, handle HTTP requests, and seamlessly connect machine learning models to web interfaces.
- **Database Management:** Gained practical experience integrating SQLite databases into Python applications to persistently store application data and prediction logs.
- **Development Tools:** Became proficient in using Google Teachable Machine for rapid prototyping, Google Stitch guidelines for professional UI design, GitHub for version control, and VS Code for efficient software development.

5.2 Machine Learning Skills

Working on consecutive, increasingly complex projects provided deep insights into the machine learning lifecycle.

- **Data Preprocessing & EDA:** Learned that the success of a model depends heavily on the quality of its training data. I mastered Exploratory Data Analysis, feature engineering, and data cleaning techniques to handle missing values and outliers effectively.
- **Model Training & Evaluation:** Gained practical experience training both Regression and Classification models. I learned to objectively compare model performance using crucial evaluation metrics such as Accuracy, Precision, Recall, and F1 Score, and apply Cross Validation to ensure robustness.
- **Advanced AI Concepts:** Deepened my understanding of Transfer Learning, allowing me to leverage powerful pretrained models like EfficientNetV2B0. Furthermore, I acquired practical knowledge of Explainable AI (XAI) using Grad-CAM, enabling the visualization of neural network decision-making processes.

5.3 Computer Vision Skills

The latter half of the internship provided a specialized focus on Computer Vision.

- **Image Processing Pipelines:** Learned the academic and practical importance of image resizing, RGB conversion, and normalization to prepare raw visual data for neural network ingestion.
- **Dataset Curation:** Understood the critical importance of dataset balancing, diversity, and augmentation in training robust image classifiers capable of generalizing to unseen data.
- **Industrial AI Deployment:** Acquired the skills to deploy TensorFlow models into production-style web applications, mastering the complete pipeline from prediction generation to confidence scoring and threshold validation.

5.4 Soft Skills Developed

In addition to technical expertise, the internship environment fostered essential soft skills required for professional software engineering.

- **Problem Solving and Debugging:** Encountering deployment issues, TensorFlow dependency conflicts, and dataset formatting errors cultivated an analytical approach to debugging and complex problem-solving.

- **Independent Learning:** The necessity to research, implement, and troubleshoot advanced technologies independently strengthened my ability to adapt and learn new tools rapidly.
- **Time Management:** Balancing multiple projects and strict deadlines leading up to the final capstone project significantly improved my ability to manage time efficiently and prioritize tasks.
- **Documentation and Presentation:** Documenting the QualiVision AI project and preparing weekly reports enhanced my technical writing and professional communication skills.
- **Professional Ethics:** Operating in a structured corporate environment instilled professional ethics, accountability, and disciplined work habits.

5.5 Challenges Faced and Resolutions

Throughout the internship, several technical challenges were encountered, each providing a valuable learning opportunity.

- **Understanding Complex Algorithms:** Initially, grasping the mathematical intuition behind machine learning algorithms was challenging. *Resolution:* This was overcome by breaking down the concepts into smaller, programmable chunks and implementing them practically on simple datasets before scaling up.
- **AutoViz Compatibility Issues:** While exploring automated EDA, version conflicts arose between AutoViz, Matplotlib, and Seaborn. *Resolution:* I resolved this by systematically debugging the Python environment, isolating dependencies, and ensuring compatible library versions were installed.
- **TensorFlow Dependency Conflicts:** During the deployment of the Smart Cam Classifier, the standard TensorFlow Keras loader failed to deserialize legacy models exported from Teachable Machine. *Resolution:* After extensive research and debugging, the issue was resolved by integrating the `tf_keras` compatibility package.
- **Frontend-Backend Integration:** Connecting the sophisticated QualiVision AI Flask backend to the dynamic, real-time frontend dashboard presented timing and data-formatting challenges. *Resolution:* This was addressed by carefully designing the REST API responses in standardized JSON formats and rigorously testing endpoint communication using debugging tools.

6. CONCLUSION

The six-week internship at Nestsoft Technomaster Pvt. Ltd. was a transformative educational experience that successfully bridged the gap between theoretical classroom learning and practical industrial application. By providing a structured, intensive immersion into the world of Artificial Intelligence and Machine Learning, the internship facilitated a profound expansion of my technical capabilities and professional confidence.

The learning journey was meticulously designed, beginning with foundational Python programming and API integrations, progressing through rigorous Data Science workflows, and culminating in advanced Deep Learning and Computer Vision applications. The successful completion of multiple intermediate projects—including the AI Translator, Text-to-Speech system, Exploratory Data Analysis modules, and Machine Learning predictive models—built the technical scaffolding necessary for complex software engineering.

The pinnacle of this internship was the successful development of the final capstone project, "QualiVision AI – AI Based Live Tomato Freshness Detection System." This enterprise-grade application synthesized all the skills acquired during the training period. By integrating a sophisticated EfficientNetV2B0 convolutional neural network with a robust Flask backend, an SQLite database, Explainable AI (Grad-CAM) visualizations, and a professional industrial dashboard, the project demonstrated the immense practical value of Artificial Intelligence in automating and optimizing industrial operations.

Beyond technical proficiency in Python, TensorFlow, Keras, OpenCV, and web development, the internship cultivated vital professional skills, including systematic debugging, independent research, technical documentation, and time management. It provided firsthand insight into the challenges of software deployment, dependency management, and real-world system integration.

In conclusion, the internship at Nestsoft Technomaster Pvt. Ltd. has been an invaluable stepping stone in my academic and professional journey. It has not only solidified my understanding of AI application development but has also equipped me with the practical, industry-ready skills required to pursue a successful career in Artificial Intelligence, Machine Learning, Computer Vision, and Software Engineering.

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